

A Reproducible Open-Set Evaluation Protocol

Why closed-set accuracy must not be reported as authentication capability

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2026-07-29

Abstract

RF fingerprinting (RFFI), the practice of identifying a physical device from manufacturing-tolerance artifacts in its transmitted waveform, is increasingly introduced alongside impressive headline numbers claiming that deep learning beats classical signal processing. This paper deals with one question that must be answered before any such number is taken at face value: does the number actually answer the question we care about? We show the weight of that question with one concrete, measured case from a public-dataset lineage. Under a protocol that selects the best-performing combination after seeing held-out results, the same model lineage reports AUROC 0.9994, a number that reads as near-perfect. But the metric that authentication actually needs to answer, the fraction of legitimate attempts accepted at a fixed low false-accept rate (TAR@FAR), falls to 0.8765 for the exact same fused-packet condition, and collapses further to 0.2235 once the evaluation is redone under a strict protocol with a worst-case 1:N impostor and a hard split between the devices used for threshold calibration and the devices used for evaluation. All three numbers are measured, not fabricated, and yet they point to different conclusions. What produces this gap is not the model. It is the rigor of the evaluation protocol.

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1 Abstract

RF fingerprinting (RFFI), the practice of identifying a physical device from manufacturing-tolerance artifacts in its transmitted waveform, is increasingly introduced alongside impressive headline numbers claiming that deep learning beats classical signal processing. This paper deals with one question that must be answered before any such number is taken at face value: does the number actually answer the question we care about? We show the weight of that question with one concrete, measured case from a public-dataset lineage. Under a protocol that selects the best-performing combination after seeing held-out results, the same model lineage reports AUROC 0.9994, a number that reads as near-perfect. But the metric that authentication actually needs to answer, the fraction of legitimate attempts accepted at a fixed low false-accept rate (TAR@FAR), falls to 0.8765 for the exact same fused-packet condition, and collapses further to 0.2235 once the evaluation is redone under a strict protocol with a worst-case 1:N impostor and a hard split between the devices used for threshold calibration and the devices used for evaluation. All three numbers are measured, not fabricated, and yet they point to different conclusions. What produces this gap is not the model. It is the rigor of the evaluation protocol.

This paper does three things. First, it defines why verification and classification are different problems that require different metrics. Second, it shows, with measured cases, the two concrete failure paths by which closed-set accuracy gets misread as evidence of authentication capability: circularity and information leakage. Third, it presents a reproducible three-tier evaluation protocol (A/B/C) that explicitly manages the independent split axes of physical transmitter, receiver, capture day, environment, and packet burst; treats a worst-case 1:N impostor scenario, where an unregistered device attempts to pass as any registered identity, as the standard evaluation target; and requires confidence intervals to be re-estimated at the physical device or session level rather than at the packet level. The conclusion is simple. It is not the tier or architecture of the model but the protocol under which it was evaluated that decides whether a reported number can be trusted.

2 Why the Evaluation Protocol Decides the Outcome

RF fingerprinting identifies a physical device from the reproducible traces that manufacturing tolerances in the analog transmit chain, oscillator phase noise, ADC/DAC nonlinearity, power-amplifier response curves, leave in the radiated waveform. It is attractive precisely because it extracts an identity signal from the physical layer alone, without embedding an additional credential, and has been proposed for rogue-device detection on factory wireless networks and industrial IoT gateway authentication alike. The problem is that most of these proposals cite, as evidence, an accuracy figure for a different question: how accurately the system names which device, among a closed list of known candidates, sent a given signal. Access control faces a different question. When a device that is not on the registered list attempts to impersonate a registered identity, can that attempt be rejected at a low false-accept rate? High accuracy on the closed question guarantees nothing about the answer to this open one.

The starting point of this paper is showing that this gap is not an abstract worry. Measuring the same trained-embedding lineage three times, under three different protocols, leads to three different conclusions. The first measurement fuses fifty packets and selects, after seeing the results, the best of eighteen combinations spanning six packet counts and three scoring methods on the held-out evaluation set; it reports AUROC 0.9994. Given that this combination was chosen after the fact rather than pre-registered, and that capture days were pooled across both enrollment and evaluation, this number likely measures not “fifty independent authentication decisions” but “signal-to-noise-boosted matching within a single capture session.” The second measurement computes, for the same selected combination, the metric authentication actually needs (TAR@FAR=1e-3); with post-hoc selection still in effect, it comes to 0.8765. The third measurement assumes a worst-case impostor that attempts to pass as any registered identity, completely separates the devices used for threshold calibration from the devices used for evaluation, and recomputes TAR@FAR=1e-3 for the same fifty-packet fusion condition (K=50); it falls to 0.2235. If the first number is quoted as reported, the system reads as a near-perfect authenticator. What the third number actually says is that, at the same false-accept target, nearly four out of five legitimate users are rejected.

Because these three numbers measure different metrics (AUROC and TAR@FAR), it would not be precise to draw a single straight line between them and say “the model got this much worse.” What this paper wants to stress is different: citing the first number as evidence of authentication capability without recognizing that it does not sit on the same coordinate system as the second and third is itself the misreading. Later sections return to these three numbers and spell out concretely why they occupy different coordinate systems, and which protocol actually corresponds to deployment conditions.

3 Verification and Classification Are Different Problems

Classification asks “which of the N registered devices sent this signal,” a task whose defining premise is that the set of correct candidates is already finite and known. Verification asks a different question: “is the signal that claims identity X actually X , or is it something else that was never registered,” a binary judgment whose defining premise is that inputs entirely absent from the candidate list are always present in the evaluation stream. Access control, authentication, and rogue-device detection are all verification problems. However high classification accuracy is, without measuring the rate at which an unregistered input is mistakenly accepted as a registered identity, one cannot know whether the system is fit for access control.

This distinction must show up directly in metric choice. Closed-set accuracy (accK) only checks whether the answer was correct among registered candidates, and is meaningful only when reported alongside chance level ($1/\text{number of registered devices}$). Because this metric never handles unregistered inputs at all, it says nothing about verification performance. Measuring verification performance requires five families of metrics.

The Equal Error Rate (EER) is the error rate at the operating point where the rate of wrongly rejecting a legitimate attempt equals the rate of wrongly accepting an unregistered one. It summarizes overall separability in a single number before any threshold has been fixed.

$$\text{EER} = \text{FAR}(\tau^*) = \text{FRR}(\tau^*), \quad \text{where } \tau^* \text{ satisfies } \text{FAR}(\tau) = \text{FRR}(\tau)$$

AUROC (area under the ROC curve) is the integral, across all thresholds, of the probability that a genuine attempt’s score exceeds an unknown attempt’s score. It represents overall discriminative power before a threshold has been chosen.

$$\text{AUROC} = P(s(x_{\text{genuine}}) > s(x_{\text{unknown}}))$$

TAR@FAR(τ) is the operating point deployment actually needs. One fixes a target false-accept rate (FAR) first, then reads off how many legitimate attempts are accepted (TAR, true accept rate) at that threshold. AUROC can be high while TAR at a specific FAR is low, and this paper’s central warning sits exactly at that point.

$$\text{TAR@FAR} = \gamma : \quad \text{TAR}(\tau_\gamma), \quad \text{where } \tau_\gamma \text{ satisfies } \text{FAR}(\tau_\gamma) = \gamma$$

OSCR (open-set classification rate) combines classification and verification in a single curve. At each false-accept level, it plots the fraction of accepted, non-rejected legitimate attempts that were also correctly classified to their registered identity. It penalizes both a system that verifies correctly but classifies wrong, and one that classifies correctly but fails to reject unregistered inputs.

Finally, calibration metrics, ECE (expected calibration error) and the Brier score, are needed. Most verification systems output a continuous confidence score rather than a binary decision, and that score is used as a risk-fusion layer alongside other signals. When a model reports “90 percent confident,” one must confirm that the prediction is in fact correct roughly 90 percent of the time before that score can safely be fused with other signals.

$$\text{ECE} = \sum_{b=1}^B \frac{|n_b|}{N} |\text{acc}(b) - \text{conf}(b)| \quad (B \text{ equal-width bins, default } B = 15)$$

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

How confidence intervals are computed matters as much as which metric is chosen. RFFI-style experiments typically evaluate signals from a few dozen physical devices, cut into thousands to tens of thousands of packets. The true independent unit here is the physical device (or capture session), not the packet. Packets from the same device and the same session are correlated. Bootstrapping a confidence interval by resampling tens of thousands of packets makes the interval artificially narrow. The true sample size is the number of devices, usually eight to twenty-four, and the confidence interval must be re-estimated by resampling whole devices (or sessions) at a time. Skipping this distinction creates the illusion that sampling variability is far smaller than it actually is.

4 The Trap of Circularity and Leakage: One Signal, Two Different Truths

Closed-set accuracy gets misread as authentication capability along two main paths. The first is a metric swap, reporting classification accuracy and citing it as if it were verification performance. The second is using the correct verification metric but on a data split that leaks information, so the number measures a task far easier than actual deployment conditions. Both are shown below with measured cases.

The first trap: an impressive closed-set accuracy, a bleak authentication operating point. Laid side by side, results from the same experiment on a public-dataset lineage make this trap plain. Under a relatively easy condition, eight registered devices, the same receiver, the same capture session, closed-set accuracy (accK) is 82.0 percent, well above chance level ($1/8 = 12.5$ percent), an impressive figure. Yet the metrics that authentication actually needs, measured on the exact same experiment and the exact same embedding, tell a different story. AUROC remains high at 0.881, but TAR@FAR=1% is 20.8 percent and TAR@FAR=0.1% is just 2.9 percent. Citing the 82.0 percent closed-set figure alone makes this system sound usable for authentication, but fixing FAR at 0.1 percent means 97 out of 100 legitimate users are rejected. Measuring the same lineage under a harder condition, a real cross-receiver split where the receiver is swapped, closed-set accuracy itself drops to 50.4 percent, and TAR@FAR=0.1% stays at 2.7 percent. Closed-set accuracy differs sharply between the two conditions (82.0 percent versus 50.4 percent), yet neither condition recovers a usable hard-authentication operating point. Closed-set accuracy is itself a metric that swings heavily with protocol difficulty, and no value it takes speaks for verification performance.

The second trap: circularity, a fingerprint that memorized the receiver rather than the device. Running a separate task on the same feature set, guessing which of twelve receivers received a given signal, at a chance level of 1 in 12 (8.3 percent), yields an accuracy of 96.4 percent. Device-identification accuracy on the exact same feature set was 92.1 percent, lower than receiver-identification accuracy. That is, this feature set memorizes receiver identity better than device identity. This mechanically explains why performance collapses when the receiver is swapped. Switching to a different feature set, one known to depend less on the receiver, reduces this leakage: receiver-identification accuracy falls to 83.1 percent and device-identification accuracy sits at 83.5 percent, a 13.3-percentage-point reduction in leakage, yet 83 percent receiver information remains overwhelmingly present. What this experiment shows is that a large fraction of an apparently correct “device identification” result can actually be “receiver identification.” Splits that enroll and evaluate on the same receiver never surface this circularity; it hides disguised as performance.

We now return to the three numbers promised in the introduction. AUROC 0.9994, reported for the best-of-eighteen combination chosen after seeing the held-out results, already falls to 0.8765 when the metric authentication actually needs (TAR@FAR=1e-3) is computed for the exact same combination. Assuming a worst-case impostor that attempts to pass as any registered identity, and completely separating the devices used for threshold calibration from the devices used for evaluation, recomputing the same fused-packet condition (K=50) drives

it down to 0.2235. The optimistic sweep’s TAR@FAR=1e-3 (0.8765) and the worst-case protocol’s identical metric (0.2235) compare the exact same metric at the exact same packet count. The gap between them is still nearly fourfold. The only variable producing that gap is not the model. It is the evaluation protocol.

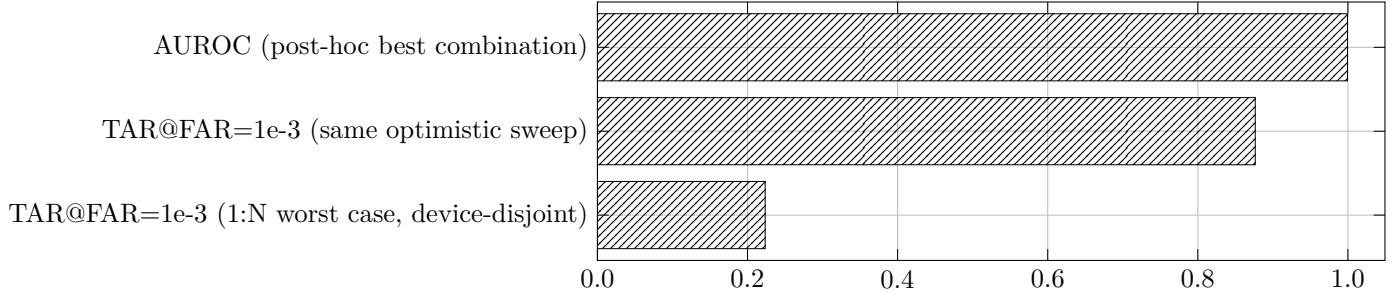


Figure 1: Gap between an optimistic and a strict protocol

It is also worth confirming that this gap is not a fluke of one particular dataset. A dedicated open-set verification method reported in the literature, using structural anchoring and rejection-oriented alignment on the same public-dataset lineage, reports OSCR 0.958 and FPR@TPR90 0.047 under a single-decision condition. Our own measurements under the same condition give OSCR between 0.326 and 0.622 and unknown-rejection rates between 10.8 percent and 29.4 percent. The gap between our own single-packet condition and this comparison point shows, once more, that open-set verification is genuinely difficult and that no single impressive headline number is enough to clear that difficulty.

5 A Recommended Protocol: A/B/C

All of the cases above share one cause. Information that leaks in RFFI/receiver verification almost always originates from one axis shared between training and evaluation. A reproducible evaluation manages the following six axes explicitly, and any experiment should document how each one was handled.

Split axis	What it is	Why it leaks if unmanaged
Transmitter / device instance	The object the fingerprint targets	Packets from the same device split across train/eval make memorization look like generalization
Device model / chipset	Model-specific spurious signal	Model-specific artifacts mix with the fingerprint
Receiver (rx chain)	Antenna, RF front end, ADC	Receiver characteristics blend into the fingerprint, becoming “memorized the receiver”
Capture day / session	Temperature drift, clock drift, environment	Same-day captures share correlated noise
Environment / location	Indoor/outdoor, distance, multipath	Environment-specific channel characteristics entangle with the fingerprint
Packet / burst group	Consecutive packets from one capture	Adjacent packets share correlated channel and timing, so random splitting effectively splits within one capture

One rule among these admits no exception: never split the same continuous capture (burst) across training and evaluation. Packet-level random splitting inflates performance artificially because adjacent packets share

channel and timing correlation. At minimum, whichever axis (device, receiver, or day) the experiment is meant to test must be an explicit hold-out.

Reporting the same experiment at three levels of how strictly these six axes are managed exposes the gap between “lab numbers” and “deployment numbers.” Every experiment should report at least one tier explicitly labeled, ideally A and B together, and C before productization.

Protocol A (research baseline). Uses the same receiver and the same or adjacent capture day, holding out only unregistered devices. Registered devices are part of the enrollment procedure; unregistered devices never appear in training or enrollment. This protocol should only be used to check the ceiling of a backbone or scoring method, never cited as evidence of product performance, and must always carry the caveat that it holds only under this condition.

Protocol B (practical open set). Separates registered and unregistered devices exactly per the open-set definition, keeps the same receiver, but separates the capture day/environment at evaluation time from training time. Unregistered data is never used for threshold decisions or model selection. If information about unregistered devices leaks into hyperparameter or threshold tuning, open-set performance is optimistically inflated. Thresholds are chosen only on the validation split of registered devices.

Protocol C (product grade). Completely holds out the actual receiver that will be deployed. Data from that receiver is used only for enrolling registered devices, never for model training, hyperparameter tuning, or threshold calibration. The deployment-time capture day and environment are also held out, simulating the real-world condition that the deployment environment differs from the training environment. Registered devices are only allowed an enrollment step on the new receiver, which mirrors the real product procedure of “enrolling an existing device on a new receiver.” Evaluation defaults to a worst-case 1:N impostor scenario: assuming that every unregistered device attempts to impersonate whichever registered identity it is closest to, and measuring the pass rate of that best-case impersonation attempt. This scenario produces the only number honest enough to cite in a product or defense brief.

Measuring under Protocol C’s worst-case condition also reveals that fusing more packets, and thereby accepting longer decision latency, helps somewhat but is not a complete fix. The table below re-measures the same system while increasing the number of fused packets (K).

K (fused packets)	TAR@FAR=1e-3	TAR@FAR=1e-4
1	0.16%	0.01%
20	6.79%	1.60%
50	22.35%	13.11%
100	43.33%	22.00%

Even at K=100, four out of five legitimate users are still rejected at the strictest target (FAR=1e-4). Looking at AUROC alongside this condition, AUROC at K=20 remains high at 0.863, while TAR@FAR=1e-3 at the same K is only 6.8 percent. This paper’s central warning, that a high AUROC and a usable authentication operating point are not the same thing, is reconfirmed here as well.

Protocol C also reveals that separating capture days is a harsher condition than swapping receivers. Under the same worst-case 1:N design, when training and evaluation are at the same point in time, AUROC is 0.9515 and closed-set identification accuracy is 88 percent. Moving evaluation to a capture two to three days in the future drops AUROC to 0.7981 and closed-set identification accuracy to 47 percent. Under the same design, swapping only the receiver dropped AUROC by an amount observed between 0.605 and 0.722, smaller than the degradation from capture-day separation. This means that in real deployment, time-related factors such as temperature drift and clock offset can be a bigger threat than hardware swaps, and re-enrollment interval should be treated as a core product design variable.

Before closing this section, a checklist for writing up a new experiment report is worth stating explicitly. State, in one line each, how each of the six split axes was managed (or explicitly note “not applicable” with a reason). Label which protocol (A/B/C) was used, in a table. Report closed-set accuracy, AUROC, AUPRC, OSCR, EER, TAR@FAR (at FAR 10%/1%/0.1%), FPR@TPR (at TPR 90%/95%), ECE, and Brier score in full, to prevent

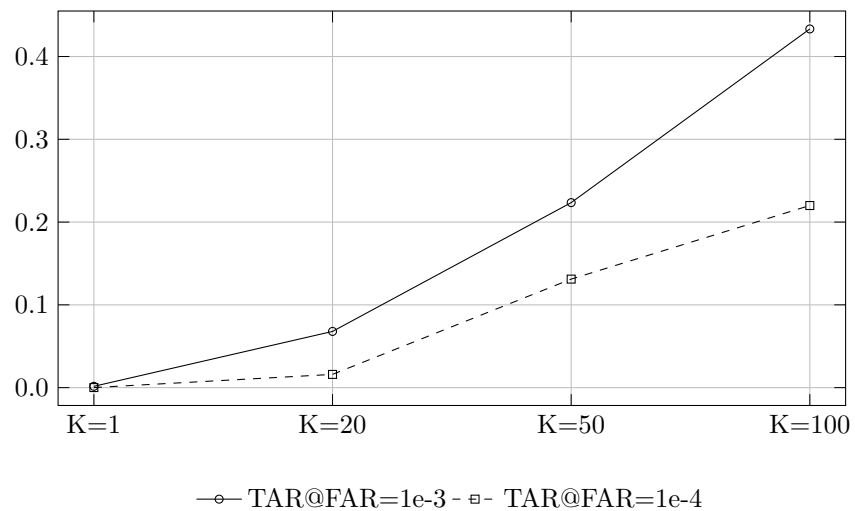


Figure 2: Fusing more packets

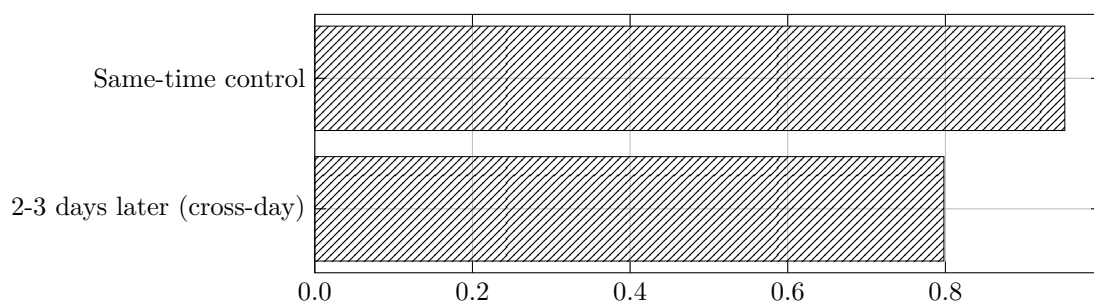


Figure 3: Cross-day separation is harsher than swapping receivers

cherry-picking a favorable subset. State whether confidence intervals are bootstrapped at the device/session level rather than the packet level. And confirm that unregistered data never entered threshold decisions or model selection.

6 Limitations

We are explicit about what this paper does not measure. Robustness under active spoofing or replay attacks is not addressed; whether high-fidelity replayed signals are indistinguishable from the baseline remains an open question requiring separate verification. Performance at enrolled-device populations much larger than tested here (tens to hundreds of thousands of devices) is out of scope. As the registered population grows, the probability that some unregistered device happens to be close to some registered identity increases, so it is safer to treat even the worst-case numbers reported here as, if anything, optimistic relative to real large-scale deployment. Point estimates from small samples (eight to twenty-four known devices) carry real sampling variability, which is not a computational error but a genuine property of small-sample estimation. Finally, the six-axis split rule and A/B/C protocol presented here were extracted from the specific task family of RFFI/receiver verification; carrying them over to other biometric or behavioral authentication tasks requires re-identifying the split axes appropriate to each task.

7 Reproducibility

The core contribution of this paper is the protocol and metric definitions, not a specific model, so the unit of reproducibility is the protocol, not the model. A researcher aiming to reproduce these results can follow the procedure below directly. First, before touching a dataset, confirm the metadata for each of the six axes (transmitter, device model, receiver, capture day/session, environment, packet/burst group), and explicitly hold out at least the axis the experiment is meant to test. Second, enforce burst-id-level group splitting in the split script so that packets from the same capture never appear on both sides of training and evaluation. Third, fix thresholds only on the validation split of registered devices, and expose unregistered data only at the final evaluation stage. Fourth, compute and report closed-set accuracy, AUROC, AUPRC, OSCR, EER, TAR@FAR, FPR@TPR, ECE, and the Brier score in full, and re-estimate confidence intervals by bootstrapping at the device (or session) level. Fifth, when reporting Protocol C, assume that every unregistered device constructs its best possible impersonation attempt against every registered identity, and compute TAR@FAR from the pass rate of that best attempt. These five steps are not tied to any particular vendor or backbone, and can be reproduced directly on public open-set RF fingerprinting datasets. The three-step gap cited in this paper, from AUROC 0.9994 through TAR@FAR=1e-3 of 0.8765 down to 0.2235, is exactly what this procedure reproduces.

8 References

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2. INRIA PLA I/Q RF Fingerprinting Dataset. Zenodo record 18268648. License: CC BY 4.0 (commercial use allowed).
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5. The definitions of the open-set verification metrics used here (EER, AUROC, TAR@FAR, OSCR, ECE, Brier score) and the device/session-level bootstrap confidence-interval procedure follow a fixed internal evaluation protocol specification (unpublished), applied to the public datasets in references 1-4 above for the measurements reported here.